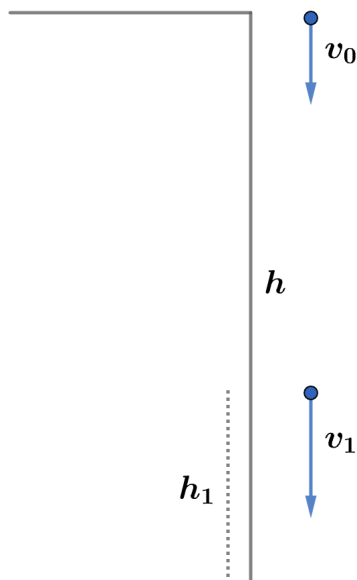


2017 F=ma Exam: Problem 17

Kevin S. Huang



We first find the velocity v_1 of the object when it is $h_1 = 60$ m above the ground. Since it takes $t = 1$ s for it to reach the ground from there, we have

$$h_1 = v_1 t + \frac{1}{2} g t^2$$
$$v_1 = \frac{h_1}{t} - \frac{1}{2} g t = \frac{60 \text{ m}}{1 \text{ s}} - \frac{1}{2} (10 \text{ m/s}^2) (1 \text{ s}) = 55 \text{ m/s}$$

Now we conserve energy between height h and height h_1 ,

$$\frac{1}{2} m v_0^2 + m g h = \frac{1}{2} m v_1^2 + m g h_1$$

Solving for the initial speed v_0 ,

$$v_0^2 + 2 g h = v_1^2 + 2 g h_1$$
$$v_0 = \sqrt{v_1^2 - 2 g (h - h_1)} = \sqrt{(55 \text{ m/s})^2 - 2 (10 \text{ m/s}^2) (180 \text{ m} - 60 \text{ m})} = 25 \text{ m/s}$$

so the answer is B.